

Queueing-Based Rate Allocation and Routing in Small Networks

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Abstract—Classical queueing models provide a theoretical solution to allocating service and transmission rates in routers and switches, guaranteeing stability and low delay. Nonetheless, this project is concerned with exploring practical simulations of these models in the context of a small network. To commence, we will calculate theoretical derivations for service and transmission rates in canonical models (in our case, we will work with M/M/1, D/D/1 and M/M/1/N). Furthermore, we will explore the influence of performance objectives such as stability, delay minimization, and loss minimization on optimal rate settings. Commencing with the small network simulation, we will implement a multi-hop routed testbed, enforcing analytically derived rate limits at each node. Lastly, we will compare real-world performance with baselines such as equal rate allocation.

Keywords—component, formatting, style, styling, insert

I. INTRODUCTION

Server to server communication relies mainly on packet switching, wherein the information sent is divided between many packets which follow their own route to the destination. Furthermore, these packets rely on an existing network infrastructure based on routers and servers to navigate to their destination. This network ensemble relies on the simultaneous implementation of numerous systems, including routing, congestion control, header processing, data processing, queue control, encoding, data encryption, etc. In our present work, we will focus on queue control, and how the various models on queue behavior function.

More concretely, with this work we set out to verify whether three types of queues following Kendall's notation have the theoretical steady-state number of packets, average number of packets, and processing time per packet that their equations predict. This demonstration serves to reinforce our confidence in present queueing infrastructure. In addition, we have graphed the time evolution of the number of packets and the processing time between packets before reaching the steady state.

Following our study, we have found that

II. RELATED WORK

The theoretical and mathematical behavior of the three queue models that pertain to the present work has been studied and developed extensively [1] [2] [3] [4].

Nevertheless, the present investigation has relied primarily on the course material for the Purdue ECE 54700 course as a base on which to build the relevant queue simulations. This work is not intended to be a review of the existing literature on queueing theory, but rather an exploration of how said theory holds up in real world case studies. Former research

works with simulations of queueing behavior include Wu et al. [5], Núñez-Martínez & Mangues-Bafalluy (WiNTECH) [6], Monowar et al. [7], Mat Nawi & Yusof → [8] and Faheem [9].

III. APPROACH

Queues can be described with Kendall's notation, with four parameters describing the packet arrival distribution, the packet departure distribution, the number of servers / queues, and the queue size / buffer. In our theoretical exposition, we will derive equilibrium equations for the three aforementioned queues (M/M/1, D/D/1 and M/M/1/N).

A. Queueing-theoretic derivations

- M/M/1

Models a single server queue with an infinite buffer. Furthermore, M/M stands for memoryless / Markov arrivals and departures, meaning that future arrivals are independent of past arrivals, because the process has no “memory” of past arrival times. More concretely, arrivals and departures follow Poisson distributions, with their consequent exponential service times. Arrivals $\sim \text{Pois}(\lambda)$ have a mean of λ and a variance of λ , with the time between packets following an $\text{Exp}(\lambda)$ distribution, with a mean of $1/\lambda$ and a variance of $1/\lambda^2$. Similarly, departures follow the same formulas but with a different μ rate.

The Poisson probability mass function is the following:

$$f(\lambda, x) = (\lambda^x \exp(-\lambda)) / x!$$

Furthermore, the exponential distribution follows the probability distribution function below:

$$f(\lambda, x) = \lambda \exp(-\lambda x)$$

One important property of Poisson distributions which can be leveraged in queueing theory derivations is PASTA (Poisson arrivals see time averages). In effect, this means that the expected number of packets in the system seen by an arriving packet is the same as the expected number of packets. Counterintuitively, this is not true for all distributions.

We define the following functions:
 $p_n(t)$: Probability of n packets in the system at time t .

p_n : limit of $p_n(t)$ as t goes to infinity (steady-state probability of having n packets in the system).

To derive the steady-state equations, we observe what happens when the number of packets at time t is n , and t is increased by an infinitesimal Δt :

$$P(\text{no increase of } n) = P(0 \text{ arrivals, } 0 \text{ departures}) + P(1 \text{ arrival, } 1 \text{ departure}) + P(k \text{ arrivals, } k \text{ departures}) = (1 - \lambda\Delta t)(1 - \mu\Delta t) + \lambda\Delta t\mu\Delta t + o(\Delta t) = (1 - (\lambda + \mu)\Delta t) + o(\Delta t).$$

$$P(\text{increase of } n \text{ by } 1) = P(1 \text{ arrival, } 0 \text{ departures}) + P(k+1 \text{ arrivals, } k \text{ departures}) = \lambda\Delta t(1 - \mu\Delta t) + o(\Delta t) = \lambda\Delta t + o(\Delta t).$$

$$P(\text{decrease of } n \text{ by } 1) = P(0 \text{ arrivals, } 1 \text{ departure}) + P(k-1 \text{ arrivals, } k \text{ departures}) = \mu\Delta t(1 - \lambda\Delta t) + o(\Delta t) = \mu\Delta t + o(\Delta t).$$

Other cases have an infinitesimal ($o(\Delta t)$) probability. Therefore:

$$pn(t+\Delta t) = pn(t) * P(\text{no increase of } n) + pn+1(t) * P(\text{increase of } n \text{ by } 1) + pn-1(t) * P(\text{decrease of } n \text{ by } 1) = pn(t) * (1 - (\lambda + \mu)\Delta t) + pn+1(t) * \lambda\Delta t + pn-1(t) * \mu\Delta t + o(\Delta t)$$

$$(pn(t+\Delta t) - pn(t)) / \Delta t = -(\lambda + \mu) pn(t) + \lambda pn-1(t) + \mu pn+1(t) + o(\Delta t)$$

As $dpn(t)/dt = 0$ in the steady-state case and $\lim \Delta t \rightarrow 0$:

$$(\lambda + \mu) pn = \lambda pn-1 + \mu pn+1$$

$$\text{Base case } (n=0): p_0 \lambda = p_1 \mu \Rightarrow p_1 = (\lambda/\mu) p_0$$

The same argument can show that $p_2 = (\lambda/\mu) p_1 = (\lambda/\mu)^2 p_0$, and therefore: $pn = (\lambda/\mu)^n p_0 = \rho^n p_0$, where $\rho = \lambda/\mu$.

$$\sum_{n=0}^{\infty} pn = 1 \Rightarrow \sum_{n=0}^{\infty} \rho^n p_0 = 1$$

$$p_0 = 1 / (\sum_{n=0}^{\infty} \rho^n) = 1 - \rho, |\rho| < 1$$

$$pn = \rho^n (1 - \rho)$$

One important observation is that if the arrival rate λ is bigger than the departure rate μ , $pn(t)$ will tend towards infinity since more packets enter the system than those that leave.

When a packet arrives to the queue, the expected waiting time is:

$$E[T] = E[W_q] + 1/\mu, \text{ where } W_q \text{ is the time for all packets ahead to be served.}$$

$$E[T] = E[N_a]/\mu + 1/\mu, \text{ where } N_a \text{ is the number of packets in the system seen by the arriving packet.}$$

From the PASTA property, we know that $E[N_a] = E[N]$, and from Little's law, we know that $E[N] = \lambda E[T]$.

$$E[T] = \rho E[T] + 1/\mu$$

$$E[T] = 1/\mu(1 - \rho) = 1/(\mu - \lambda), \text{ where } \mu > \lambda.$$

$$E[N] = \sum_{n=0}^{\infty} n pn = (1 - \rho) \sum_{n=1}^{\infty} n \rho^n = \rho(1 - \rho)/(1 - \rho)^2 = \rho/(1 - \rho)$$

- Finite-buffer M/M/1/N

Same as the M/M/1 queue, but with a finite buffer of size N instead of an infinite buffer.

$$pn = \rho^n p_0 \text{ as before, but } 0 \leq n \leq N.$$

$$p_0 = 1 / (\sum_{n=0}^N \rho^n) = (1 - \rho) / (1 - \rho^{N+1})$$

$$pn = (\rho^n (1 - \rho)) / (1 - \rho^{N+1}), 0 \leq n \leq N$$

As before, $\rho > 1$, but here we can calculate what happens for the edge case when $\rho=1$.

$$p_0 \sum_{n=0}^N \rho^n = 1 \Rightarrow p_0 = 1 / (N+1)$$

$$E[T] =$$

$$E[N] = \sum_{n=0}^N n pn = (1 - \rho) \sum_{n=1}^N n \rho^n / (1 - \rho^{N+1}) =$$

$$\rho(1 - (N+1)\rho^N + N\rho^{N+1}) / (1 - \rho^{N+1})(1 - \rho)$$

- D/D/1

Single server queue with an infinite buffer. The key difference with the M/M/1 model is that arrivals and departures follow a deterministic, or degenerate rate, instead of a rate with a random probability.

B. Small multi-hop routed testbed approach

To build the network simulation, we have used a Python object-oriented procedure.

IV. RESULTS AND DISCUSSION

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V. CONCLUSION AND FUTURE WORK

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VI. TEAM PARTICIPATION

- Pérez: abstract, introduction, related work, queueing-theoretic derivations, baseline code for the simulation.
- Muraleedharan:
- Parikh:

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